



**COURSE:PROBABILITY & STATISTICS
MT-331**

**Lecture 5
Part (3)**

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Expected value

Let X be a discrete random variable with probability function $p(x)$. Then the *expected value* of X , $E(X)$,

is defined to be

$$E(X)^r = \sum X^r P(X = x_i) \quad r=1,2,3,4$$

$$\mu = E(x) = \sum x P(X = x_i) \quad E(X^2) = \sum X^2 P(X = x_i)$$

PROPERTIES:

- The expected value of a constant is a constant itself. i.e $E(k) = k$
- $a E(x) = E(ax)$
- If X & Y two random variable then $E(X \pm Y) = E(X) \pm E(Y)$
- if X & Y two independent variable then $E(XY) = E(X)E(Y)$
- $V(X) = E(X^2) - (E(X))^2$

Example: Discrete random variable x has prob. Distribution as under.

Then find (i) $E(4x + 6)$ (ii) $V(2x + 3)$

- $E(4x + 6) = E(4X) + E(6)$
 $= 4E(X) + 6$

$$E(X) = \sum X P(X)$$

$$= \frac{5}{3}$$

$$= 4\left(\frac{5}{3}\right) + 6$$

$$E(4x + 6) = 38 / 3$$

X	P (X)	X. P(X)
0	1 / 3	0
1	1 / 4	1 / 4
2	1 / 12	2 / 12
3	1 / 8	3 / 8
4	1 / 6	4 / 6
5	1 / 24	5 / 24

$$\sum P(X) = 1$$

$$\sum X P(X) = 5 / 3$$

Example: Discrete random variable x has prob. Distribution as under.

Then find (i) $E(4x + 6)$ (ii) $V(2x + 3)$

$$\bullet E(X) = \sum X P(X) = \frac{5}{3}$$

$$V(2X + 3) = ?$$

$$\begin{aligned} V(2X + 3) &= V(2X) + V(3) \\ &= 4V(X) + 0 \end{aligned}$$

$$V(X) = E(X^2) - (E(X))^2$$

$$\begin{aligned} E(X^2) &= \sum X^2 P(X) \\ &= 121 / 24 \end{aligned}$$

$$V(X) = 121 / 24 - (5 / 3)^2 = 163 / 72$$

$$V(2X + 3) = 4(163 / 72) = 9.055$$

X	P (X)	X ² P(X)
0	1 / 3	0
1	1 / 4	1 / 4
2	1 / 12	4 / 12
3	1 / 8	9 / 12
4	1 / 6	16 / 6
5	1 / 24	25 / 24
$\sum P(X) = 1$		121 / 24

Example:

Three fair coins tossed at a time .Draw probability distribution table where X represent the no of HEAD .Calculate mean & variance ,skewness & kurtosis of the pdf table?

x	$f(x_i)$	$x.f(x_i)$
0	$1/8$	0
1	$3/8$	$3/8$
2	$3/8$	$6/8$
3	$1/8$	$3/8$
Total	1	$12/8$

Mean

$$\mu = E(x)$$

$$E(X) = \sum X P(X)$$

$$\mu = 12 / 8$$

x	$f(x_i)$	$x^2 \cdot f(x_i)$
0	$1/8$	0
1	$3/8$	$3/8$
2	$3/8$	$12/8$
3	$1/8$	$9/8$
Total	1	$24/8$

$$\mu = E(x) = \sum xP(X = x_i) = 12/8$$

Variance: $V(X) = E(X^2) - (E(X))^2$

$$E(X^2) = \sum X^2 P(X = x_i)$$

$$= 24/8$$

$$V(X) = 24/8 - (12/8)^2$$
$$= 0.75$$

x	$f(x_i)$	$x^2 \cdot f(x_i)$	$x^3 \cdot f(x_i)$	$x^4 \cdot f(x_i)$
0	1/8	0	0	0
1	3/8	3/8	3/8	3/8
2	3/8	12/8	24/8	48/8
3	1/8	9/8	27/8	81/8
Total	1	24/8	54/8	132/8

$$E(X) = 12/8 = \mu \quad \sqrt{\beta_1} = \frac{\mu_3}{(\mu_2)^{3/2}}$$

$$E(X_2) = 24/8 = \mu'_2$$

$$E(X_3) = 54/8 = \mu'_3$$

$$E(X_4) = 132/8 = \mu'_4$$

$$\begin{aligned} \mu_3 &= E(X_3) - 3E(X_2)E(X_1) + 2(E(X_1))^3 \\ &= (54/8) - 3(24/8)(12/8) + 2(12/8)^3 \\ \mu_3 &= 0 \end{aligned}$$

$$\begin{aligned} V(X) &= \mu_2 = E(X_2) - (E(X))^2 \\ V(X) &= \mu_2 = 0.75 \end{aligned}$$

$$\beta_2 = \frac{\mu_4}{(\mu_2)^2}$$

$$\mu_4 = E(X_4) - 4E(X_3)E(X_1) + 6E(X_2)(E(X_1))^2 - 3(E(X_1))^4$$

$$= 132/8 - 4(54/8)(12/8) + 6(24/8)(12/8)^2 - 3(12/8)^4$$

$$= 21/16$$

$$E(X) = 12/8 = \mu$$

$$E(X_2) = 24/8 = \mu'_2$$

$$E(X_3) = 54/8 = \mu'_3$$

$$E(X_4) = 132/8 = \mu'_4$$

$$\begin{aligned}\sqrt{\beta_1} &= \frac{\mu_3}{(\mu_2)^{3/2}} \\ &= \frac{0}{(0.75)^{3/2}} = 0\end{aligned}$$

$$\begin{aligned}\beta_2 &= \frac{\mu_4}{(\mu_2)^2} \\ &= \frac{21}{16} \\ &= \frac{21}{(0.75)^2} \\ &= 2.33\end{aligned}$$

1. The probability distribution of a random variable X is given below.

$X = x_i$	1	2	3	4	5
$P(X = x_i)$	k	$2k$	$3k$	$4k$	$5k$

Find the value of k , mean and variance of X .

Sol: We know that $\sum P(x_i) = 1$

$$\Rightarrow k + 2k + 3k + 4k + 5k = 1$$

$$\Rightarrow 15k = 1$$

$$\Rightarrow k = \frac{1}{15}$$

Mean: $\mu = \sum X_i P(x_i)$

$$\Rightarrow \mu = 1(k) + 2(2k) + 3(3k) + 4(4k) + 5(5k)$$

$$\Rightarrow \mu = 55k$$

$$\Rightarrow \mu = 55 \left(\frac{1}{15} \right)$$

$$\therefore \text{Mean } (\mu) = \frac{11}{3}$$

Variance

$$\sigma^2 = \sum x_i^2 P(x_i) - \mu^2$$

$$\Rightarrow \sigma^2 = (1)^2(k) + (2)^2(2k) + (3)^2(3k) + (4)^2(4k) + (5)^2(5k) - \left(\frac{11}{3}\right)^2$$

$$\Rightarrow \sigma^2 = k + 8k + 27k + 64k + 125k - \frac{121}{9}$$

$$\Rightarrow \sigma^2 = 225k - \frac{121}{9}$$

$$\Rightarrow \sigma^2 = 225 \left(\frac{1}{15}\right) - \frac{121}{9}$$

$$\Rightarrow \sigma^2 = 15 - \frac{121}{9}$$

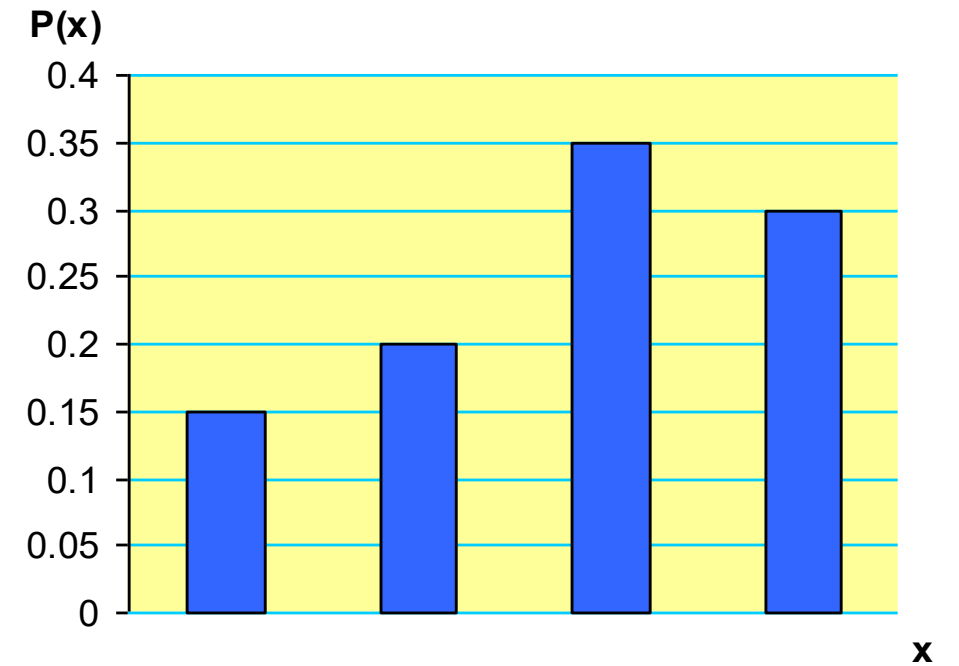
$$\text{Variance } (\sigma^2) = \frac{14}{9}$$

The following table lists the probability distribution of the number of breakdowns per week for a machine based on past data.

Breakdowns per week	0	1	2	3
Probability	.15	.20	.35	.30

- a) **Present this probability distribution graphically.**
- b) **Find the probability that the number of breakdowns for this machine during a given week is**
 - i. **exactly 2**
 - ii. **0 to 2**
 - iii. **more than 1**
 - iv. **at most 1**

x	$P(x)$
0	.15
1	.20
2	.35
3	.30
$\Sigma P(x) = 1.00$	



(b)

Breakdowns per week	0	1	2	3
Probability	.15	.20	.35	.30

- i. $P(\text{exactly 2 breakdowns}) = P(x = 2) = .35$
- ii. $P(0 \text{ to } 2 \text{ breakdowns}) = P(0 \leq x \leq 2)$
 $= P(x = 0) + P(x = 1) + P(x = 2)$
 $= .15 + .20 + .35 = .70$
- iii. $P(\text{more than 1 breakdown}) = P(x > 1)$
 $= P(x = 2) + P(x = 3)$
 $= .35 + .30 = .65$
- iv. $P(\text{at most one breakdown}) = P(x \leq 1)$
 $= P(x = 0) + P(x = 1)$
 $= .15 + .20 = .35$
- v. $P(\text{at least three breakdown}) = P(x \geq 3) = 0.30$

IN THE LAST LECTURE YOU LEARNT

Chapter no 3

• Discrete Random Variable

- Discrete Probability Distribution
- Graphical Representation of a Discrete Probability Distribution
- Mean, Standard Deviation and Coefficient of Variation of a Discrete Probability Distribution
- Distribution Function of a Discrete Random Variable.

OBJECTIVE

Chapter no 5

• Discrete Probability Distribution

- Binomial Distribution
- Graphical Representation of a Binomial Probability Distribution
- Mean, Standard Deviation of a Binomial Probability Distribution

OBJECTIVE

Chapter no 5

• Discrete Probability Distribution

- Multinomial Distribution
- Negative Binomial Distribution
- Geometric Distribution
- Poisson Distribution

Binomial Distribution

A Bernoulli trial can result in a success with probability p & a failure with probability $q=1-p$. then the probability distribution of the binomial random variable X , the no of successes in n independent trial is

$$b(x; n, p) = \binom{n}{x} p^x q^{n-x} \quad x = 0, 1, 2, \dots, n$$

where

n = total number of trials

p = probability of success

$q = 1 - p$ = probability of failure

x = number of successes in n trials

$n - x$ = number of failures in n trials

$$\begin{aligned} \sum (P + q)^n &= 1 \\ \text{Mean} = \mu &= E(X) = np \\ \text{variance} &= V(X) = npq \\ \text{Standard deviation} &= \sqrt{npq} \end{aligned}$$

BERNOULLI TRIAL –

Events that meet the following 3 conditions:

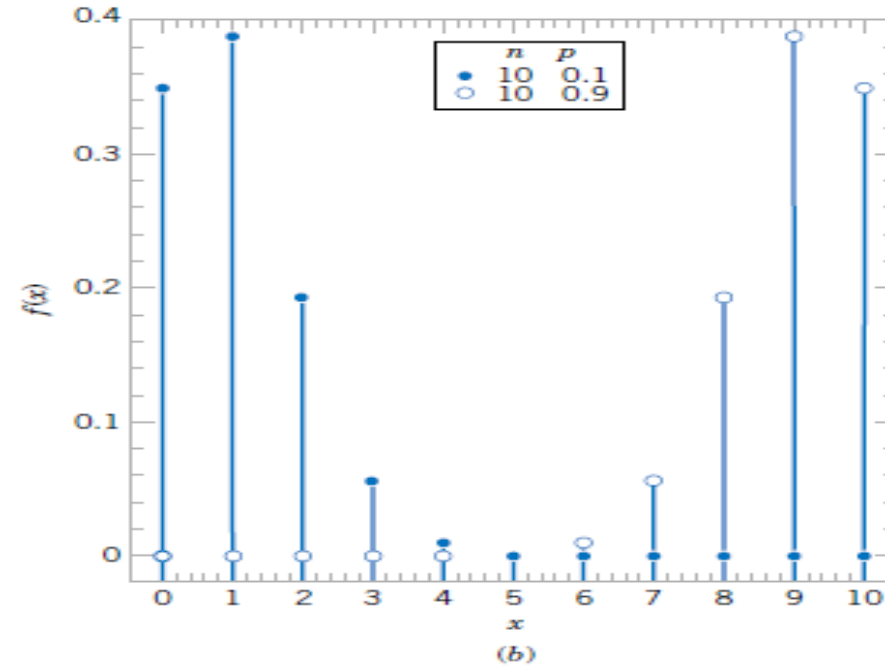
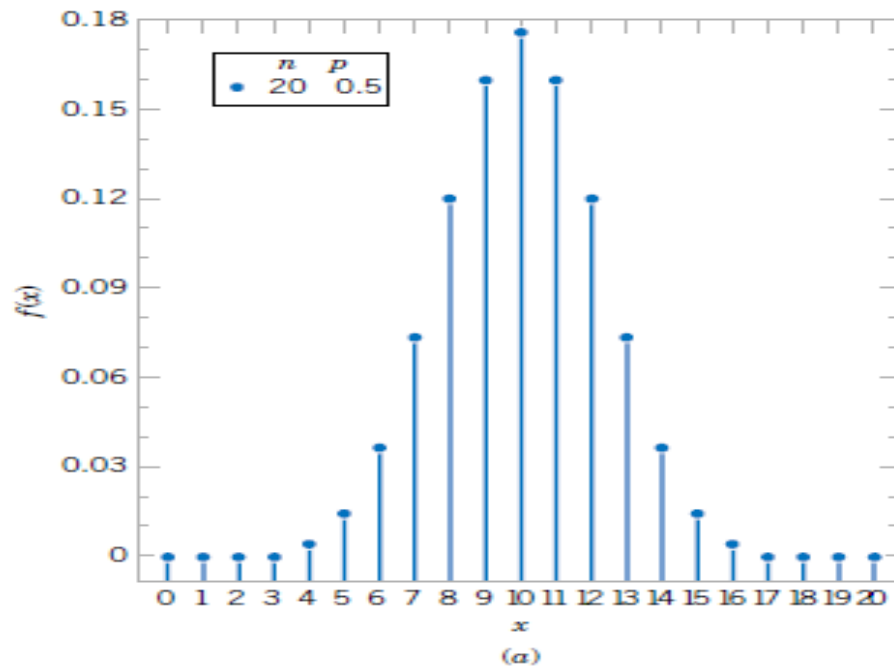
- There are only two outcomes.**
- The probability of success is constant.**
- The trials are independent.(Must meet all 3 conditions above.)**

EXAMPLES: flipping heads on a coin, finding a defect, finding a prize, making a basket, getting a hit.

Conditions of a Binomial Experiment

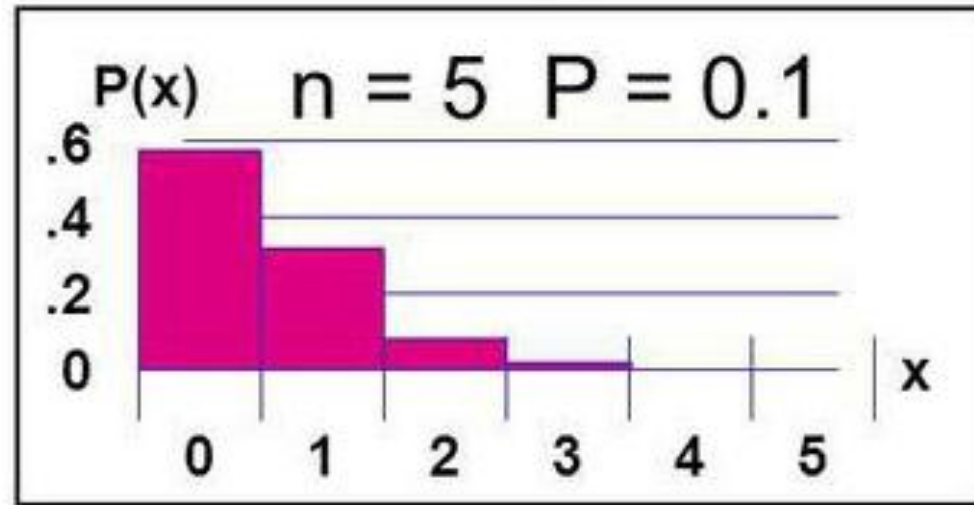
1. **There must be a fixed number of trials.**
2. **Each trail has only two possible outcomes. These outcomes can be considered as either success or failure.**
3. **The probabilities of a success must remain the same for each trial.**
4. **The trials are independent.**
5. **In binomial distribution there are two parameters n & p .**
6. **Variance is always less than its mean.**
7. **When n is very large and p or q is very small, Binomial distribution tend to Poisson Distribution**
8. **When n is very large and p or q are not very small, Binomial distribution tend to Normal Distribution**

- 9- When p and q are equal i.e. $p = q = \frac{1}{2}$, B.D. is a symmetrical distribution.
When $p < \frac{1}{2}$, its skewness is positive and
when $p > \frac{1}{2}$, its skewness is negative.

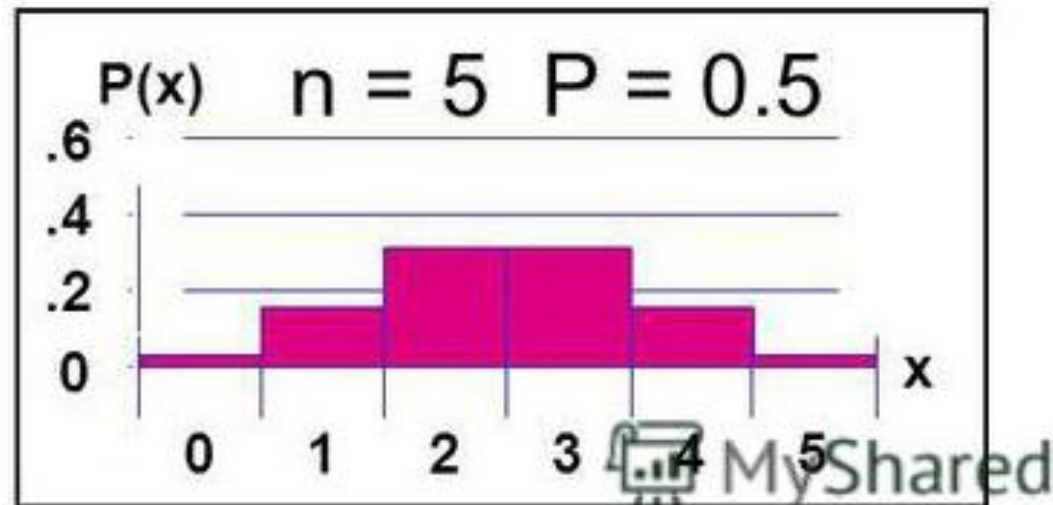


- The shape of the binomial distribution depends on the values of P and n

- Here, $n = 5$ and $P = 0.1$



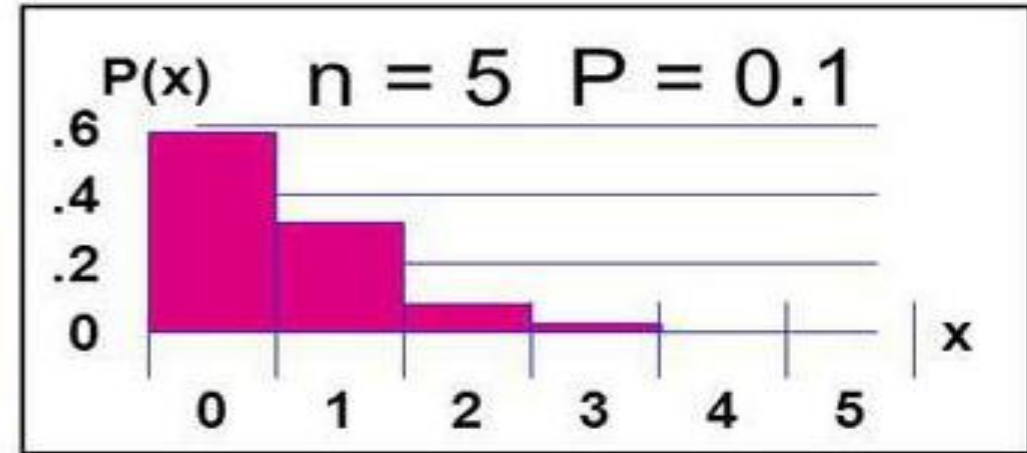
- Here, $n = 5$ and $P = 0.5$



Examples

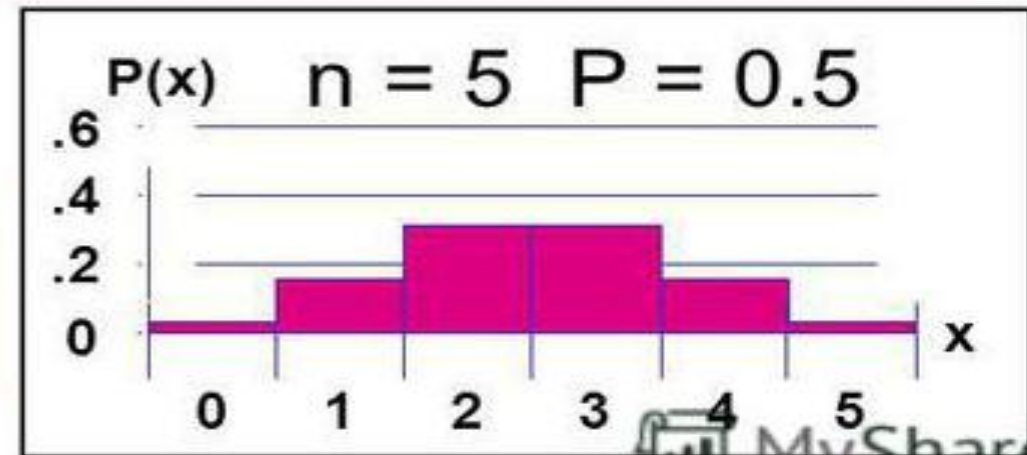
$$\mu = nP = (5)(0.1) = 0.5$$

$$\sigma = \sqrt{nP(1-P)} = \sqrt{(5)(0.1)(1-0.1)} = 0.6708$$



$$\mu = nP = (5)(0.5) = 2.5$$

$$\sigma = \sqrt{nP(1-P)} = \sqrt{(5)(0.5)(1-0.5)} = 1.118$$



$$P(x) = \frac{n!}{x! (n-x)!} P^x (1-P)^{n-x}$$

$P(x)$ = probability of x successes in n trials,
with probability of success P on each trial

x = number of 'successes' in sample,
($x = 0, 1, 2, \dots, n$)

n = sample size (number of trials
or observations)

P = probability of "success"

Example: Flip a coin four
times, let x = # heads:

$$n = 4$$

$$P = 0.5$$

$$1 - P = (1 - 0.5) = 0.5$$

$$x = 0, 1, 2, 3, 4$$

- **5.2** Twelve people: are given two identical speakers to listen to for differences, if any. Suppose that these people answered by guessing only. Find the probability that three people claim to have heard a difference between the two speakers?

$$b(x; n, p) = c_x^n p^x q^{n-x} \quad x = 0, 1, 2, \dots, n$$

Sol

$$\begin{aligned} n &= 12 \\ p &= 0.5 \\ q &= 1 - p = 0.5 \\ x &= 3 \end{aligned}$$

$$\begin{aligned} b(3; 12, 0.5) &= \frac{12!}{3!(12-3)!} 0.5^3 0.5^{12-3} \\ &= 0.0537 \end{aligned}$$

- **5.4** In a certain city district the need for money to buy drugs is stated as the: reason for 75% of all thefts. Find the probability that among the next 5 theft cases reported in this district,
(a) exactly 2 resulted from the need for money to buy drugs.
(b)

$$\begin{aligned} n &= 5, \quad p = 0.75, \quad X = 2, \quad q = 0.25 \\ b(2; 5, 0.75) &= \frac{5!}{2!(5-2)!} 0.75^2 0.25^{5-2} \\ &= 0.087 \end{aligned}$$

(b) at most 3 resulted from the need for money to buy drugs.

$$(b) P(X \leq 3) = \sum_{l=0}^3 b(x; 5, 0.75)$$

$$P(X \leq 3) = b(0; 5, 0.75) + b(1; 5, 0.75) + b(2; 5, 0.75) + b(3; 5, 0.75)$$

$$P(X \leq 3) = \sum_{i=0}^3 (C_x^5)(0.75)^x(1 - 0.75)^{5-x}$$

$$P(X \leq 3) = \binom{5}{0} (0.75)^0 (0.25)^{5-0} + \binom{5}{1} (0.75)^1 (0.25)^{5-1} + \binom{5}{2} (0.75)^2 (0.25)^{5-2} +$$

$$\binom{5}{3} (0.75)^3 (0.25)^{5-3}$$

$$P(X \leq 3) = 0.3672.$$

5.5 According to Chemical Engineering/ Progress (Nov. 1990), approximately 30% of all pipework failures in chemical plants are caused by operator error. What is the probability that out of the next 20 pipework failures at least 10 are due to operator error?

Solution:

$$p = 0.30, n = 20$$

Or

$$P(X \geq 10) = 1 - P(X < 10)$$

$$= 1 - \sum_{i=0}^9 b(X; 20, 0.30)$$

$$= 1 - \sum_{i=0}^9 (C_x^{20})(0.30)^x(1 - 0.30)^{20-x}$$

You can calculate this value via table.

$$= 1 - 0.9520$$

$$= 0.0480$$

$$\begin{aligned} P(X \geq 10) &= \sum_{i=10}^{20} b(X; 20, 0.30) \\ &= \sum_{i=10}^{20} (C_x^{20})(0.30)^x(1 - 0.30)^{20-x} \\ &= 0.0480 \end{aligned}$$

Table A.1 (continued) Binomial Probability Sums $\sum_{x=0}^r b(x; n, p)$

<i>n</i>	<i>r</i>	<i>p</i>									
		0.10	0.20	0.25	0.30	0.40	0.50	0.60	0.70	0.80	0.90
20	0	0.1216	0.0115	0.0032	0.0008	0.0000					
	1	0.3917	0.0692	0.0243	0.0076	0.0005	0.0000				
	2	0.6769	0.2061	0.0913	0.0355	0.0036	0.0002				
	3	0.8670	0.4114	0.2252	0.1071	0.0160	0.0013	0.0000			
	4	0.9568	0.6296	0.4148	0.2375	0.0510	0.0059	0.0003			
	5	0.9887	0.8042	0.6172	0.4164	0.1256	0.0207	0.0016	0.0000		
	6	0.9976	0.9133	0.7858	0.6080	0.2500	0.0577	0.0065	0.0003		
	7	0.9996	0.9679	0.8982	0.7723	0.4159	0.1316	0.0210	0.0013	0.0000	
	8	0.9999	0.9900	0.9591	0.8867	0.5956	0.2517	0.0565	0.0051	0.0001	
	9	1.0000	0.9974	0.9861	0.9520	0.7553	0.4119	0.1275	0.0171	0.0006	
	10		0.9994	0.9961	0.9829	0.8725	0.5881	0.2447	0.0480	0.0026	0.0000
	11		0.9999	0.9991	0.9949	0.9435	0.7483	0.4044	0.1133	0.0100	0.0001
	12		1.0000	0.9998	0.9987	0.9790	0.8684	0.5841	0.2277	0.0321	0.0004
	13			1.0000	0.9997	0.9935	0.9423	0.7500	0.3920	0.0867	0.0024
	14				1.0000	0.9984	0.9793	0.8744	0.5836	0.1958	0.0113
	15					0.9997	0.9941	0.9490	0.7625	0.3704	0.0432
	16					1.0000	0.9987	0.9840	0.8929	0.5886	0.1330
	17						0.9998	0.9964	0.9645	0.7939	0.3231
	18						1.0000	0.9995	0.9924	0.9308	0.6083
	19							1.0000	0.9992	0.9885	0.8784
	20								1.0000	1.0000	1.0000

$$\sum_{x=0}^9 b(X; 20, 0.30) = 0.9520$$

5.6 .According to a survey by the Administrative Management Society, one-half of U.S. companies give employees 4 weeks of vacation after they have been with the company for 15 years. Find the probability that among 6 companies surveyed at random, the number that give employees 4 weeks of vacation after 15 years of employment is

- (a) anywhere from 2 to 5**
- (b) fewer than 3.**

Solution:

(a) For $n = 6$ and $p = 1/2$.

$$\begin{aligned} P(2 \leq X \leq 5) &= \sum_{i=2}^5 (C_x^6)(0.5)^x(1 - 0.5)^{6-x} \\ &= \binom{6}{2}(0.5)^2(0.5)^{6-2} + \binom{6}{3}(0.5)^3(0.5)^{6-3} + \binom{6}{4}(0.5)^4(0.5)^{6-4} + \binom{6}{4}(0.5)^4(0.5)^{6-4} \end{aligned}$$

$$= 0.875$$

$$(b) P(X < 3) = P(X \leq 2) = \sum_{i=0}^2 b(x; 6, 0.50)$$

$$= \sum_{i=0}^2 \binom{6}{x} (0.5)^x (1 - 0.5)^{6-x}$$

$$= \binom{6}{0} (0.5)^0 (0.5)^{6-0} + \binom{6}{1} (0.5)^1 (0.5)^{6-1} + \binom{6}{2} (0.5)^2 (0.5)^{6-2}$$

$$= 0.3438.$$

Table A.1 Binomial Probability Sums $\sum_{x=0}^r b(x; n, p)$

n	r	p					
		0.10	0.20	0.25	0.30	0.40	0.50
6	0	0.5314	0.2621	0.1780	0.1176	0.0467	0.0156
	1	0.8857	0.6554	0.5339	0.4202	0.2333	0.1094
	2	0.9842	0.9011	0.8306	0.7443	0.5443	0.3438
	3	0.9987	0.9830	0.9624	0.9295	0.8208	0.6563
	4	0.9999	0.9984	0.9954	0.9891	0.9590	0.8906
	5	1.0000	0.9999	0.9998	0.9993	0.9959	0.9844
	6	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

5.9 In testing a certain kind of truck tire over a rugged terrain, it is found that 25% of the trucks fail to complete the test run without a blowout. Of the next 15 trucks tested, find the probability that

- (a) from 3 to 6 have blowouts;
- (b) fewer than 4 have blowouts;
- (c) more than 5 have blowouts.

Solution:

For $n = 15$ and $p = 0.25$, we have

$$\begin{aligned} \text{(a) } P(3 \leq X \leq 6) &= \sum_{i=3}^6 \binom{15}{i} (0.25)^i (1 - 0.25)^{15-i} \\ &= \binom{15}{3} (0.25)^3 (0.75)^{12} + \binom{15}{4} (0.25)^4 (0.75)^{11} + \binom{15}{5} (0.25)^5 (0.75)^{10} + \\ &\quad \binom{15}{6} (0.25)^6 (0.75)^9 \\ &= 0.7073 \end{aligned}$$

(b) $P(X < 4) = P(X \leq 3)$

$$= \sum_{i=0}^3 (C_x^{15}) (0.25)^x (1 - 0.25)^{15-x}$$

$$= \binom{15}{0} (0.25)^0 (0.75)^{15-0} + \binom{15}{1} (0.25)^1 (0.75)^{14} + \binom{15}{2} (0.25)^2 (0.75)^{13} + \binom{15}{3} (0.25)^3 (0.75)^{12}$$

= 0.4613

(c) $P(X > 5) = 1 - P(X \leq 5)$

$$= 1 - \left(\sum_{i=0}^5 (C_x^{15}) (0.25)^x (1 - 0.25)^{15-x} \right)$$

= 1 - 0.8516

= 0.1484

Table A.1 (continued) Binomial Probability Sums $\sum_{x=0}^r b(x; n, p)$

n	r	P					
		0.10	0.20	0.25	0.30	0.40	0.50
15	0	0.2059	0.0352	0.0134	0.0047	0.0005	0.0000
	1	0.5490	0.1671	0.0802	0.0353	0.0052	0.0005
	2	0.8159	0.3980	0.2361	0.1268	0.0271	0.0037
	3	0.9444	0.6482	0.4613	0.2969	0.0905	0.0176
	4	0.9873	0.8358	0.6865	0.5155	0.2173	0.0592
	5	0.9978	0.9389	0.8516	0.7216	0.4032	0.1509
	6	0.9997	0.9819	0.9434	0.8689	0.6098	0.3036

Example 3.17. An exciting computer game is released. Sixty percent of players complete all the levels. Thirty percent of them will then buy an advanced version of the game. Among 15 users, what is the expected number of people who will buy the advanced version? What is the probability that at least two people will buy it?

Solution. Let X be the number of people (successes), among the mentioned 15 users (trials), who will buy the advanced version of the game. It has Binomial distribution with $n = 15$ trials and the probability of success

$$\begin{aligned} p &= P\{\text{buy advanced}\} \\ &= P\{\text{buy advanced} \mid \text{complete all levels}\} P\{\text{complete all levels}\} \\ &= (0.30)(0.60) = 0.18. \end{aligned}$$

Then we have

$$\mathbf{E}(X) = np = (15)(0.18) = \underline{2.7}$$

and

$$P\{X \geq 2\} = 1 - P(0) - P(1) = 1 - (1 - p)^n - np(1 - p)^{n-1} = \underline{0.7813}.$$

The last probability was computed directly by formula (3.9) because the probability of success, 0.18, is not in [Table A2](#). ◇